Closing the Usability Gap in Naked Objects

From Domain Models to Complete Applications

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Abstract

This paper proposes a conjecture about user interfaces for structured information: that nearly every screen in such an application can be described by three decisions—which part of the object graph is **visible**, whether each visible child is **layered** beside its parent or **inlined** within it, and how sibling collections are **arranged**. We call a screen described this way a **lens** over the model. If the conjecture holds, a framework can own the mechanics of those decisions (navigation, collapse, responsive switching, editing, persistence) while the developer states only the policy, as hints on the model.

Strvct is an implementation of this idea in the **Naked Objects** tradition, which derives user interfaces from domain objects and their declared behavior. It adds a compositional interaction grammar—recursive master-detail navigation, inline documents, and responsive anchored tabs—and reuses the same model metadata for persistence, exchange, and agent interfaces. A production application, **undreamedof.ai**, shows the approach is feasible at scale; the argument for its generality rests on the reader's own experience of the interfaces they already know.

CCS Concepts: • **Software and its engineering** → **Object oriented frameworks**; *Model-driven software engineering*; • **Human-centered computing** → *Interactive systems and tools.*

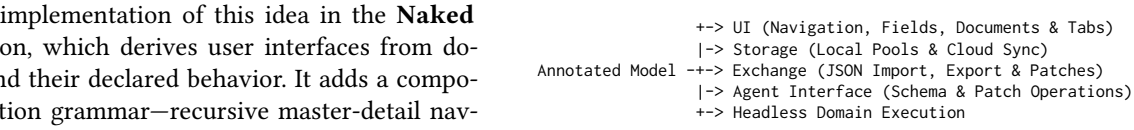
Keywords: naked objects, model-driven development, user interface generation, projection, lenses, composable primitives, AI agents, Miller columns

1 Introduction

A note on what kind of document this is. We do not offer a controlled evaluation or a measured productivity comparison. What follows is a design conjecture, an implementation that makes it concrete, and a deployed application that shows it is feasible. The evidence we ask the reader to weigh is their own: test the three-decision grammar of §3 against the information applications you know—mail clients, settings panels, issue trackers, CRMs—and see whether any screen needs a decision it cannot express. The strongest case against the conjecture would be a common, non-graphical information screen that does; we would like to hear of one. The hedges that remain in later sections mark where implementation lags the conjecture, not uncertainty about the conjecture itself.

Applications often repeat domain structure across model declarations, input controls, validation, persistence mappings, navigation, and integration schemas. A new property can require coordinated changes in several places. Shared components and schema-driven tools reduce this duplication, but application authors still commonly maintain a separate presentation structure.

Naked Objects, described by Pawson and Matthews [11], offers another approach: expose domain objects and actions through generated interfaces. Strvct retains that model-first architecture while exploring a richer set of reusable navigation and layout mechanisms.



The contribution is a compositional account of those UI mechanisms and an implementation that reuses model metadata across subsystems. The division of labor matters: the framework does not infer layouts, as early Naked Objects systems did; it implements the mechanics, and the developer states intent. Domain design, presentation decisions, and testing the resulting experience remain the developer's work—but as policy, not as separately authored screens.

2 The Usability Problem Addressed Here

Generating fields and collections is only part of generating a useful application. Users also need to retain context while traversing relationships, see related information together, and navigate on small screens. This paper focuses on three design requirements:

1. **Visible hierarchy:** Distinguish ownership, containment, and cross-references.
2. **Context-preserving navigation:** Keep the selected path legible as users move between related objects.
3. **Reusable responsive behavior:** Adapt the same navigation structure to different viewport sizes.

Forms and tables remain useful arrangements; the question is how they compose with other views. Existing model-driven systems also support customization: Apache Causeway, for example, documents UI extensions for maps and calendars [1]. Strvct's focus here is recursive composition of layered and inline views, rather than a claim that prior systems can render only forms and tables.

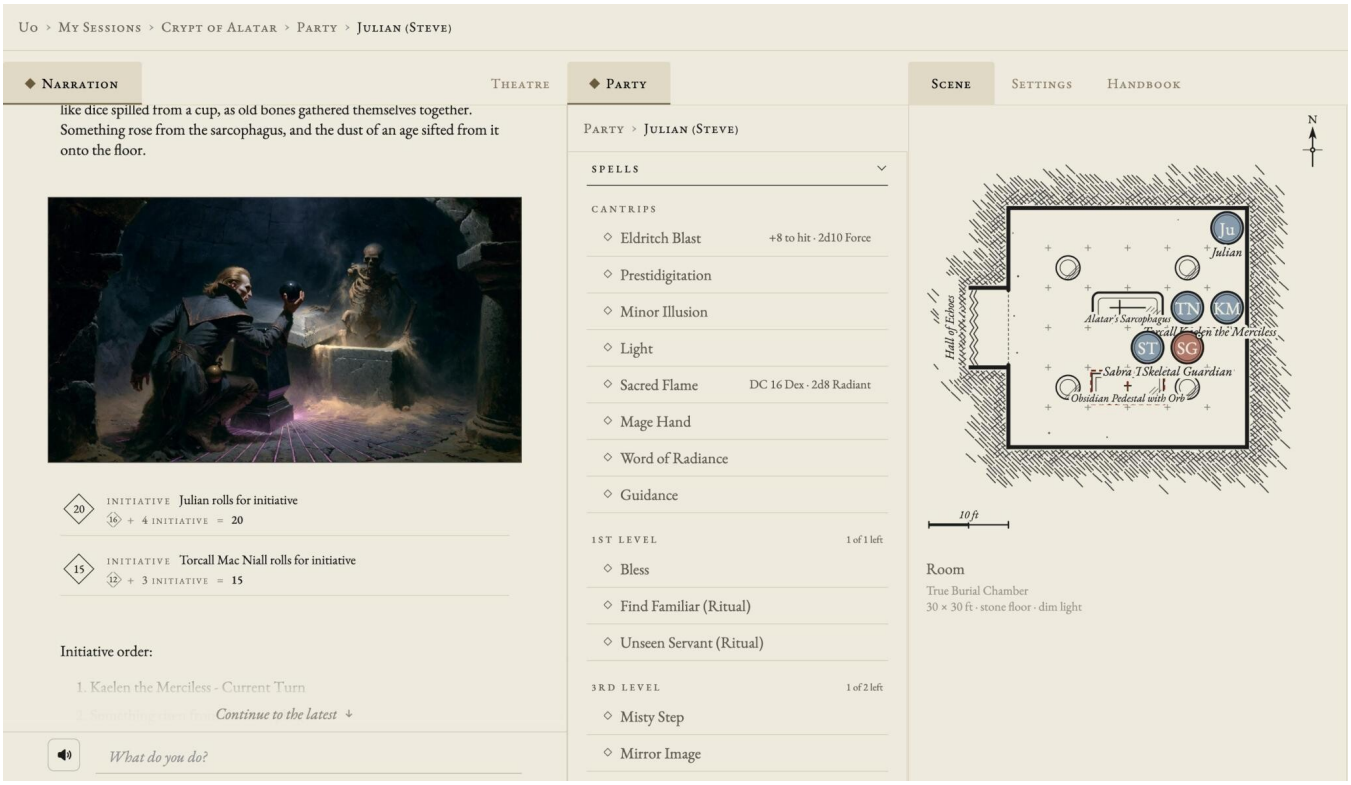


Figure 1. Production example: undreamedof.ai.

3 The Model as a Graph, the Screen as a Lens

This section states the conjecture from the Abstract precisely. Before describing the grammar in detail, it helps to picture what a generated interface is actually showing.

Imagine the domain model as a graph: a tree of ownership, with symbolic links for the references that cross it. Now imagine drawing all of it at once—every object, every property, every collection, fully expanded. This drawing is the model's complete **UI projection**. This is a conceptual projection of the UI-exposed model, not a drawing that the runtime must materialize; internal or hidden properties need not be exposed.

Every screen the user sees is a **lens** over that drawing. At first order, a lens makes only two decisions:

1. **Visibility:** Which nodes to collapse. Typically everything off the current navigation path is collapsed, leaving the path itself and the immediate children of the nodes along it.
2. **Placement:** For each visible parent→child edge, whether the child is **layered** (given its own plane beside or in front of its parent, as in Miller columns, push stacks, and tabs) or **inlined** (drawn within its parent's extent, as in documents, grouped sections, and side-by-side panes).

Layering and inlining are not modes of the whole interface but choices made edge by edge, and they compose freely. A layered column may hold an inlined document whose link opens another layer; a document section may itself be a set of anchored tabs. The mixed lens in Figure 2 inlines *Project*'s sections in one column and layers *Task 2* into the next, and even the pure inlining lens layers its link to *Person*. Because the choice lives on each edge rather than on each screen, it can change without touching the model—set by a developer's hint, or switched by the framework as space changes.

At second order, a lens decides **arrangement**: how siblings within a layer or an inlined run are laid out—their orientation, whether their properties align across siblings into a table, or whether a value maps to position, as on a timeline (§4.1). Visibility, placement, and arrangement are the three decisions of the conjecture.

This framing reduces an apparently unbounded space of screens to something tractable. Within this grammar, a screen is described by a navigation path, placements for visible edges, and arrangements for visible collections. That is a small, regular decision space—small enough for the framework to own its mechanics (the path, collapse, layering, inlining, and responsive switching) while the developer states intent through hints on the model: which edges inline,

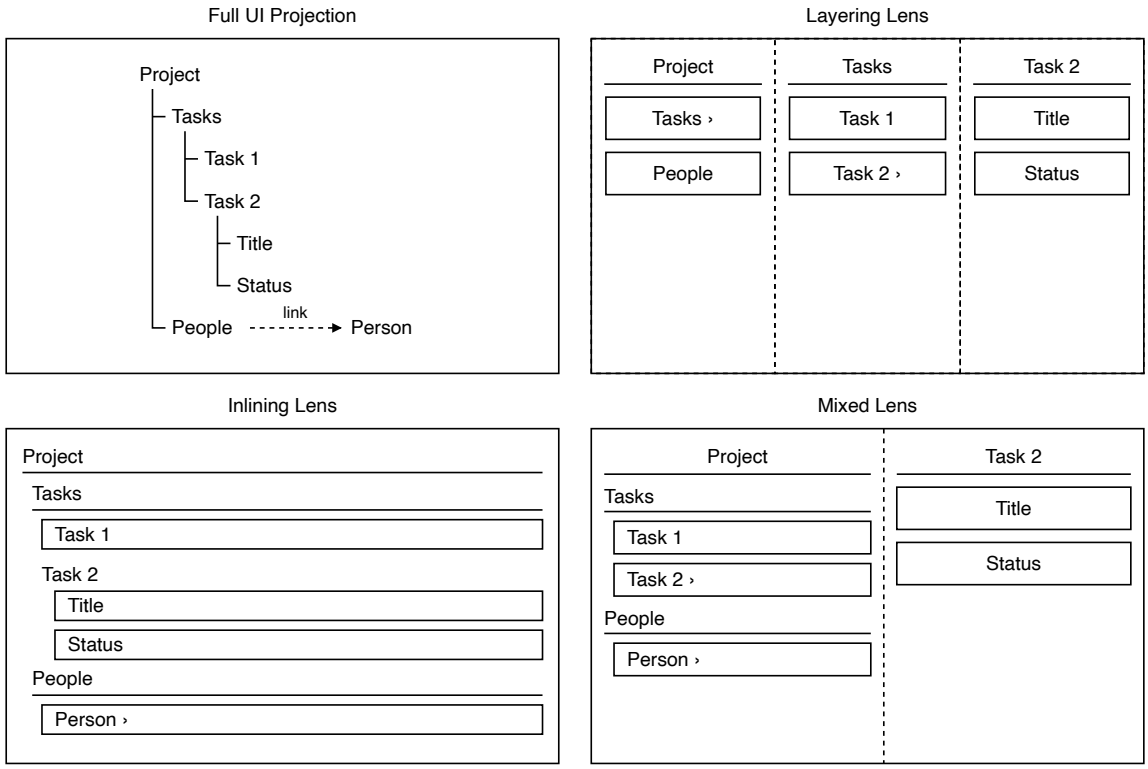


Figure 2. One Projection, Three Lenses: a small model graph as its full UI projection, and through layering, inlining, and mixed lenses.

which layer, how collections arrange. The decisions are the developer's; their implementation is the framework's.

It also explains behaviors that would otherwise have to be built screen by screen:

- **Navigation** is moving the lens. Traversing a relationship changes the path, not the model.
- **Responsive collapse** is the lens switching edges from inlined to layered as space runs out, within the bounds the hints allow. Miller columns that collapse into a push stack are still layered—in depth rather than side by side. Side-by-side panes that degrade to tabs are the same edge moving from inlined to layered.
- **Links** normally open another layer, making a cross-reference explicit. Recursive inlining also needs depth or cycle limits; layering alone does not bound the length of a navigation history.

None of these decisions is new to the literature. Furnas's fisheye views [7] formalize visibility as a degree-of-interest function over a tree; overview+detail and focus+context interfaces [3] separate or blend levels of detail much as layering and inlining do; and tree visualization distinguishes children drawn *within* their parents (inclusion) from children drawn *beside* them (adjacency) [12]. Inlining a linked target is what Nelson called transclusion [10]. Systems such as Furnas's, APT [9], and SUPPLE [8] go further and compute the presentation—by interest score, by search, or by optimization. Strvct deliberately does not. It owns the mechanism and supplies the defaults; the developer states intent through hints, much as a projectional editor lets a language designer declare the notation through which a model is edited [13].

Our use of *lens* differs from the Magic Lenses of HCI [2], which filter a local region of the screen. It shares the editable-view intuition of bidirectional programming [6], because edits flow back to the model; this analogy does not assert the formal round-trip laws of a bidirectional lens.

The UI projection is not the only one. The same graph is projected into records in object pools (the **storage projection**), into JSON for import, export, and patching (the **exchange projection**), and into a schema-and-patch tool interface for AI agents (the **agent projection**, §7.1). Each reads the same annotated model, while requiring its own framework implementation and application configuration.

4 A Narrow Design Space for Information Apps

The conjecture's scope is structured-information applications: CRMs, project management tools, mail clients, issue trackers, settings panels, and the like. The claim is that

their screens can be expressed through relationship traversal, property editing, domain actions, and semantic collection arrangements. Readers can check this against interfaces they use: a mail client is a layered column of mailboxes, a layered column of messages, and an inlined message body; a settings panel is a vertical master-detail with inlined groups; an issue page is an inlined document whose cross-references layer. A modal is a layered edge drawn in front rather than beside; a search result is a derived collection; a dashboard is a set of inlined siblings with quantitative arrangement.

What the grammar does not supply is the task-specific workflow—the triage sequence, the bulk edit, the pipeline report. Those remain application design, stated as domain actions and policies (§6.2) rather than as screens; the framework owns the interaction mechanics beneath them.

4.1 Relationship Traversal vs. Collection Arrangement

In terms of the lens (§3), we divide the user interface grammar into two distinct layers:

- **Relationship Traversal:** Moving the lens along links in the object graph (e.g., from a *Project* to its *Tasks*, then to a specific *Task*).
- **Collection Arrangement:** Displaying a set of related objects in a specific semantic layout (e.g., viewing *Tasks* as an ordered list, a Kanban board grouped by status, a table, or a timeline).

4.2 The Boundary of Automatic Generation

Generation covers the arrangements and controls the framework implements. A developer selects those mechanisms through metadata or node methods. A new arrangement or specialized interaction may require a new renderer; once implemented generically, that renderer can potentially serve multiple domain models.

Spatial meaning alone does not determine this boundary: both calendars and battle maps encode meaningful positions. The practical distinction is whether an existing renderer supports the required semantics and interactions. Creative canvases, code editors, 3D viewports, medical imaging, and specialized scientific plots often require additional rendering and interaction code.

An application can combine generated navigation, properties, and actions with such specialized surfaces. The amount of custom work depends on its tasks, not just its domain category.

Historically, the cost of that boundary was paid in human effort against the framework's view internals. Because a specialized renderer works against a fixed contract (a node's typed slots, notifications and actions) rather than wiring its own data access, it is a well-bounded task for a coding agent. Whether this narrows the boundary in practice is untested.

5 Architecture & UI Mechanics in Strvct

Strvct implements this grammar using client-side JavaScript: traversal (visibility and placement) through master-detail views and inline layouts, and arrangement through the collection views they contain. Master-detail views are its layering lens (§5.2); inline layouts are its inlining lens (§5.3). The core visual building blocks are **Tiles**, **Tile Stacks**, and **Master-Detail Views**.

5.1 Tiles and Tile Stacks

The fundamental primitive in Strvct is the **Tile**:

- **Summary Tile:** Displays a domain object with a title, subtitle, icon, and contextual actions. Selecting a summary tile opens its details in an adjacent column.

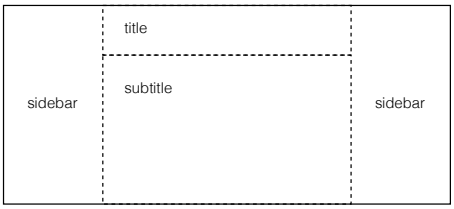


Figure 3. Summary Tile.

- **Property Tile:** Displays an individual object property (key-value pair) with direct inline editing (text inputs, numbers, toggles, date pickers) and validation feedback.

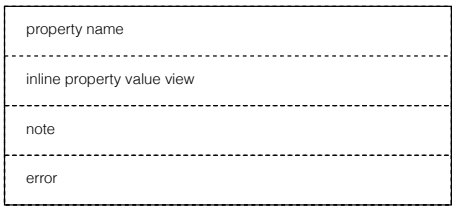


Figure 4. Property Tile.

A **Tile Stack** is a scrollable, ordered list or grid of tiles. Stacks support drag-and-drop reordering, slide-to-delete, and direct data import/export.

5.2 Recursive Master-Detail Navigation

A **Master-Detail View** pairs a master tile stack with a detail pane showing the selected object's contents. Each selection layers the chosen child in a new pane. Because detail panes can themselves contain child master-detail views, navigation composes recursively to arbitrary depth.

Key characteristics of Strvct's master-detail engine:

- **Flexible Orientation:** Each level can layout horizontally (like macOS Miller column file browsers [4]) or vertically (like nested accordion lists or settings panels).

Mechanism	Semantic mapping	Examples	Implementation scope
Recursive composition	Parent-child relationships and presentation subtrees	Master-detail navigation, inline documents, anchored tabs	Implemented; arbitrary presets such as Kanban need their own grouping and interaction policies
Cross-sibling alignment	Equivalent property slots aligned across items	Tables, spreadsheets, pivot matrices	Proposed framework extension
Value-to-space mapping	Temporal or spatial values mapped to position and extent	Timelines, Gantt charts, calendars, maps	Part of the design space; no general generated arrangement established here
Quantitative encoding	Numeric values mapped to marks or axes	Line charts, bar plots, heatmaps	Part of the design space; specialized renderers required unless supplied by the framework

Table 1. Arrangement mechanisms, the forms they yield, and their implementation scope in Strvct.

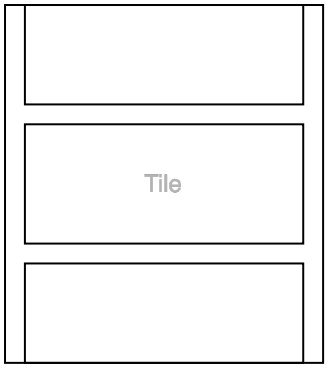


Figure 5. Tile Stack.

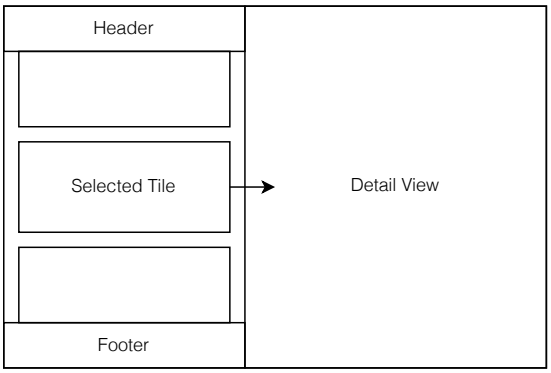


Figure 6. Master-Detail View.

- **Automatic Mobile Collapse:** When viewport space is constrained, earlier columns automatically collapse into a mobile push/pop stack equipped with interactive breadcrumbs—the same layers, stacked in depth rather than side by side. The shared navigation engine handles this collapse; application-specific content still needs responsive verification.

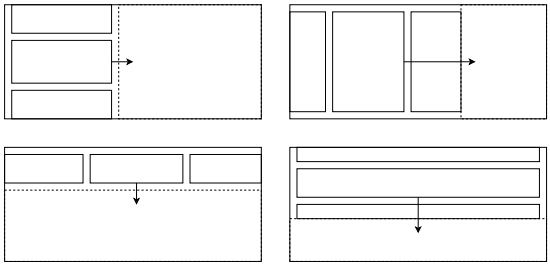


Figure 7. Flexible Master-Detail Orientations.

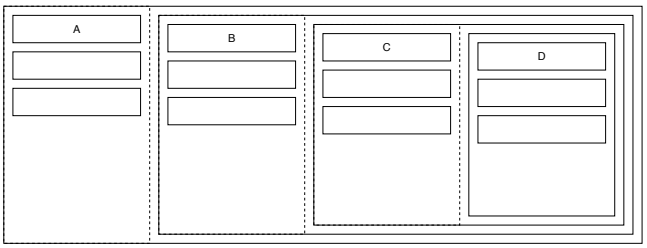


Figure 8. Expanded Viewport (Desktop).

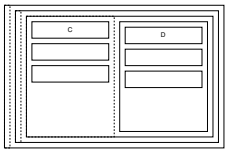


Figure 9. Collapsed Viewport (Mobile with Breadcrumbs).

- **Compositional Diversity:** Combining vertical, horizontal, and hybrid orientations expresses Miller column browsers, email clients, settings panels, chat rooms, and inspectors within a single uniform primitive.

5.3 Advanced Inline Layouts

While column traversal is the default, certain complex screens require rendering child views directly inside parent

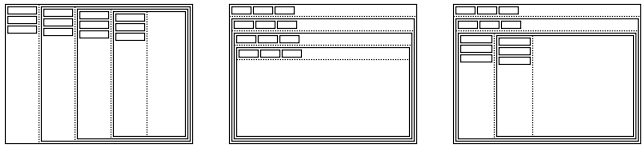


Figure 10. Vertical (left), Horizontal (center), and Hybrid (right) nesting.

tiles rather than pushing new columns. These children are inlined rather than layered. Strvct provides two inline layout mechanisms and a presentation-delegation hook:

1. **Anchored Tabs:** Renders selected children as interactive side-by-side panes or tab bars. Width arbitration automatically fits as many panes as space permits on wide desktop monitors while gracefully degrading to standard tabs on mobile.
2. **Document Inlining:** Renders a nested subtree as a continuous multi-column document with flowing sections, headers, and fields. Fields edit in place without triggering navigation transitions.
3. **Presented Nodes (Presentation Delegation):** Allows an object to delegate its presentation to a derived node (e.g., displaying a structured RPG character sheet instead of raw property slots), providing bespoke reading views without custom component templates.

6 Model Declarations & The Annotation Bridge

Developers define domain models as JavaScript classes inheriting from `SvStorableNode`. Properties are declared as **slots** enriched with metadata annotations:

```
(class Character extends SvStorableNode {

  initPrototypeSlots () {
    // Name slot: String property, persisted,
    // synced to view, editable
    {
      const slot = this.newSlot("name", "");
      slot.setSlotType("String");
      slot.setShouldStoreSlot(true);
      slot.setSyncsToView(true);
      slot.setCanInspect(true);
      slot.setIsSubnodeField(true);
      slot.setCanEditInspection(true);
    }
    // Level slot: Number property
    {
      const slot = this.newSlot("level", 1);
      slot.setSlotType("Number");
      slot.setShouldStoreSlot(true);
      slot.setSyncsToView(true);
      slot.setCanInspect(true);
      slot.setIsSubnodeField(true);
    }
    // Inventory relationship: navigable child
    {
      const slot = this.newSlot("inventory", null);
      slot.setSlotType("Inventory");
      slot.setShouldStoreSlot(true);
      slot.setFinalInitProto(Inventory);
      slot.setIsSubnodeField(true);
    }
  }

  initPrototype () {
    this.setShouldStore(true);
    // fields are UI, not stored children
    this.setShouldStoreSubnodes(false);
  }

  title () {
    return this.name() || "Unnamed Character";
  }

  subtitle () {
    return "Level " + this.level();
  }

}.initThisClass());
```

6.1 Worked Example: From a Model to a Lens

The `Character` declaration assumes an application-defined `Inventory` node class. `isSubnodeField` exposes each property as a field in the node's contents; `canInspect` also exposes it in the developer inspector. Neither persistence nor view synchronization alone makes a property visible.

With the default `nodeChildrenLayout()` value of `"column"`, selecting a character opens its fields and Inventory relationship in the next pane. Selecting Inventory traverses that relationship. On a narrow viewport, earlier navigation panes collapse and breadcrumbs retain access to the path.

To show the same contents as a document, the class can override an existing presentation hook:

```
// Methods inside Character; the stored
// properties remain the same.
nodeChildrenLayout () {
  return "inline";
}

nodeInlineDepth () {
  return 1;
}
```

If Inventory is large and should remain a navigable summary within that document, its class can override `nodeIsInlined()` to return `false`. A container that should show its children as responsive anchored tabs instead overrides `nodeSubnodesAreAnchoredTabs()` to return `true`.

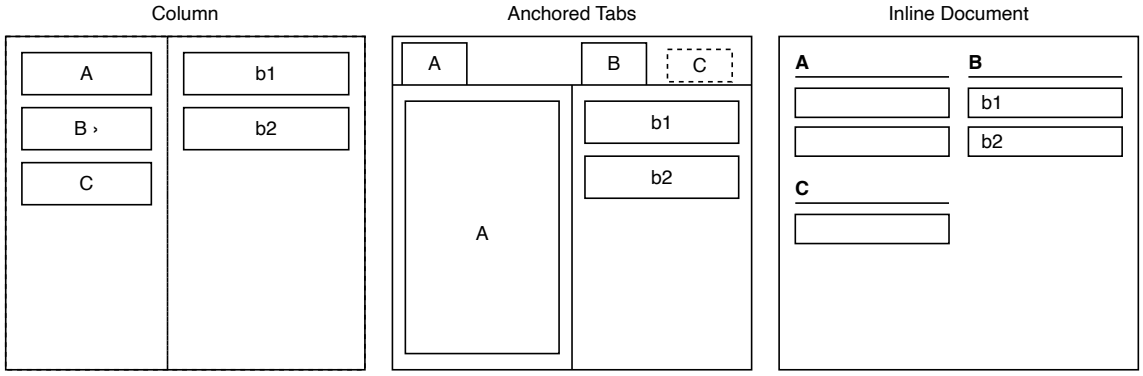


Figure 11. Comparison of Subtree Layouts: Columns, Anchored Tabs, and Inline Document Flow.

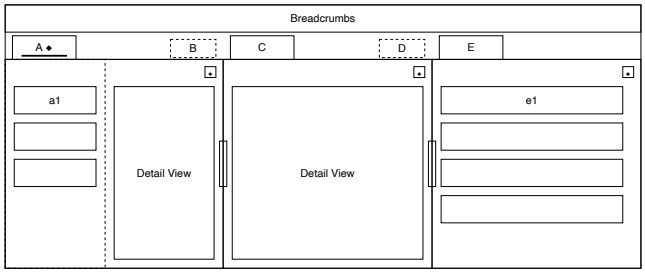


Figure 12. Anchored Tabs (Wide Screen).

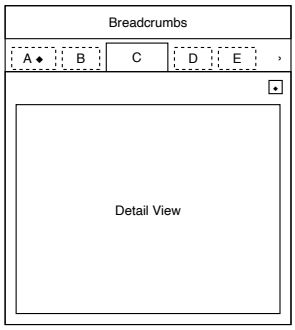


Figure 13. Anchored Tabs (Narrow Screen).

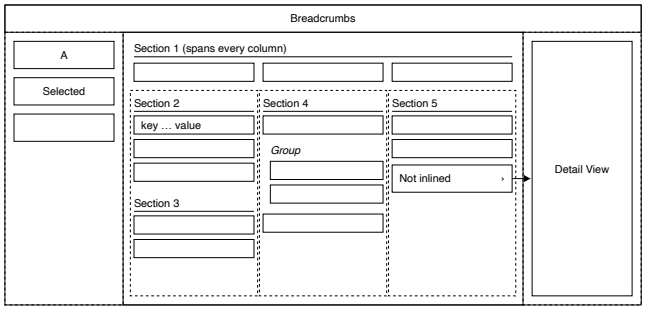


Figure 14. Inline Document Flow.

These are node-level policies interpreted by shared views, rather than separately authored screens.

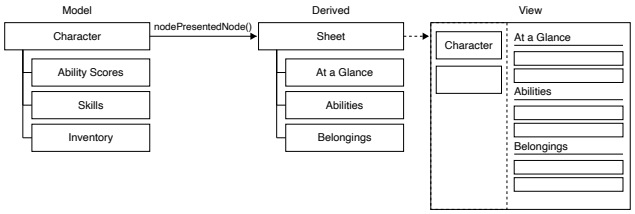


Figure 15. Presented Node Pattern.

Adding a `species` slot with the same type, visibility, persistence, and synchronization annotations as `name` makes it available to the shared field renderer and storage serializer. Labels, constraints, and presentation order still need deliberate choices. This illustrates the maintenance benefit: reuse of the field machinery rather than the elimination of presentation design.

6.2 Actions and Workflows

Domain behavior must also be exposed deliberately. An action slot names a model method, and an optional `<methodName>ActionInfo()` method supplies presentation state. For example, these members can be added to `Character`:

```
// Inside initPrototypeSlots():
{
  const slot = this.newSlot("advanceLevelAction",
                            null);
  slot.setSlotType("Action");
  slot.setIsSubnodeField(true);
  slot.setActionMethodName("advanceLevel");
}

// Methods on Character:
advanceLevel () {
  if (this.level() >= 20) {
    throw new Error(
      "Character is already at the maximum level");
  }
  this.setLevel(this.level() + 1);
```

```
}

advanceLevelActionInfo () {
  return {
    title: "Advance Level",
    isEnabled: this.level() < 20,
    isVisible: true
  };
}
```

The shared action field reads availability from the model and invokes the method; domain checks remain necessary for callers outside the UI. Slot constraints support field validation, while cross-property rules and failed operations need useful application feedback. Bulk operations and multi-step workflows require explicit domain methods and state; they do not follow automatically from navigation relationships.

6.3 The Multi-Reader Annotation Architecture

Slot annotations serve as a decoupling bridge. Each framework subsystem reads slot metadata independently without cross-system dependencies:

- **UI Layer (UI Projection):** Reads `slotType`, `canEditInspection`, and `isSubnodeField` to instantiate Property Tiles and navigate child subnodes.
- **Persistence Layer (Storage Projection):** Reads `shouldStoreSlot` to serialize records into local storage (IndexedDB) and cloud object pools.
- **Sync Scheduler:** Reads `syncsToView` to observe slot updates and post coalesced, deferred notifications to active views.
- **Agent Projection:** Reads property types and constraints to auto-generate JSON validation schemas and patch tools.
- **Localization Engine:** Reads property names and translation contexts to auto-translate labels.

```
                +-> UI Rendering (Property Tiles & Controls)
                |-> Persistence (IndexedDB & Cloud Commit)
Slot Metadata +-> Sync Scheduler (Weak Observers)
                |-> AI Agent Infrastructure (Patch Schemas)
                +-> Internationalization (Label Translation)
```

7 Shared Architectural Capabilities

Shared metadata lets framework subsystems reuse declarations. These capabilities require framework implementation, application configuration, and testing; their benefit is that much of that work is shared across generated views.

7.1 Natively AI-Operable Model

The agent projection uses slot metadata for a schema-and-patch interface, described here by two generic operation types:

1. `schema-fetch`: Retrieves the slot schema for any domain class or object instance.
2. `apply-patch`: Applies a JSON patch payload to update properties or mutate child relationships.

The number of operation types can remain constant as classes are added; schema volume, discovery work, and agent context do not. Invalid patches can return schema information for a correction attempt. A usable deployment must also define the exposed root objects, traversal and access rules, and how domain actions are invoked. A small tool vocabulary alone does not establish task success or low reasoning cost.

7.2 Headless Domain Execution and Testing

Domain classes that keep browser dependencies in the view and service layers can run under Node.js. Tests can instantiate model graphs, invoke methods, and assert slot states without browser automation. This makes domain testing cheaper; layout, focus, gestures, and browser integration still require UI tests.

7.3 Transparent Persistence and Cloud Sync

Strvct divides persistence into two unified strategies:

- **Local Object Pools:** In-memory access to loaded domain objects, with asynchronous IndexedDB persistence.
- **Content-Addressable Blob Store:** Asynchronous SHA-256 keyed storage for large binary assets (images, audio, files).

Cloud synchronization sends versioned atomic commits containing only the modified slot records. Conflicts are detected deterministically per document pool.

7.4 Shared Accessibility and Localization

Accessibility and localization are properties of the UI projection rather than projections of their own, and both are handled at the framework primitive level: translation is applied at the model-to-view boundary, as labels leave the model. Shared view primitives centralize ARIA roles, focus management, keyboard navigation, and translation hooks. Fixes in a primitive can benefit every view that uses it. Application-specific labels, reading order, custom renderers, assistive-technology behavior, and translation quality still need validation.

7.5 Live Inspector & Developer Mode

Every tile supports modifier-clicking (Option + Click) to reveal a live Property Inspector displaying all declared slots, storage flags, and validation states. Developers can toggle Developer Mode to inspect hidden internal subnodes without writing custom debugging views.

8 A Deployed Application: undreamedof.ai

Strvct was used to build **undreamedof.ai**, a commercial AI-powered virtual tabletop application for tabletop roleplaying games (D&D 5e).

Subsystem	Classes (est.)	Custom views
Character system	~30	None
Campaign hierarchy	~20	None
Session & gameplay	~25	7
AI integration & tools	~15	None
Other	—	4
Total	**~90**	**11**

Table 2. Reported application inventory of undreamedof.ai.

The application uses framework views for character sheets, inventory managers, spellbooks, campaign trees, settings, chat logs, and inspectors. Its specialized surfaces are mostly graphical or highly interactive: a 3D WebGL dice roller, a tactical battle map, the session's chat stage, and print previews.

The class counts are the author's estimates. The custom-view count comes from the source tree, counting the application's view and tile classes as of November 2026. Class counts are not screen counts, and the inventory is offered as an existence proof that the grammar covers a substantial application, not as a coverage ratio. The more informative evidence would be a worked example of one screen whose presentation hint grew into a presented node or a new renderer under product pressure, so a reader can see what "policy-sized" looks like in practice; that is planned for a future revision, along with a reproducible comparison of representative tasks on completion rate, time, navigation errors, and mobile usability.

9 Comparison with Related Paradigms

These categories overlap: component applications can derive forms from schemas, and low-code platforms can infer interfaces from data. Strvct's distinguishing commitment is to use the annotated object graph as the primary source for navigation, editing, and several non-UI subsystems. Table 3 describes architectural defaults, not measured productivity differences.

10 Limitations & Future Work

Several limitations and questions remain:

1. **Renderer Coverage:** Canvas editing, CAD, video sequencing, and 3D gaming typically need specialized renderers and interaction policies. The generation boundary depends on the controls and arrangements available in the framework (§4.2).
2. **Ecosystem Isolation:** To support content-addressable bundling and zero-build client loading, Strvct uses a custom resource loader rather than standard npm/Webpack/Vite tooling.
3. **Aligned Grid Arrangements (Tables & Spreadsheets):** While inline document flow and anchored tabs are production-proven, cross-sibling column alignment (making row tiles align into full spreadsheets) is currently a proposed framework extension.
4. **Extreme Collection Scaling:** Lazy view instantiation keeps rendering lightweight, but virtualized viewport scrolling for mega-collections (10^5 or more items) remains an active area of optimization.
5. **Evaluation and Workflows:** As stated in §1, comparative usability, accessibility, and development effort have not been measured, and complex multi-step workflows are stated as domain actions rather than derived from the grammar. The planned follow-up is described in §8.
6. **User-Chosen Lenses:** Because placement and arrangement are per-edge hints rather than authored screens, users could in principle reshape their own lenses—inlining a section they always want open, or layering one they rarely need. This is possible today through the live inspectors (§7.5), but not every hint is persisted; a supported, persistent user-facing control is future work. Engelbart's NLS gave users *viewspecs*—how many levels to show, how many lines per item—for much the same purpose [5].

11 Conclusion

Strvct demonstrates a model-first approach in which shared views interpret an annotated object graph. Its implemented navigation grammar combines recursive master-detail panes, inline documents, responsive anchored tabs, and presentation delegation. The lens model explains how these mechanisms can compose without a separately authored component tree for every screen.

The deployed application shows the grammar covers a substantial product; quantitative coverage, development cost, and comparative usability remain open, and the conjecture itself is offered for the reader to test against the interfaces they know. Tables with cross-sibling alignment and other semantic arrangements remain extensions rather than established capabilities of this implementation.

The practical promise is reduced duplication: developers maintain domain behavior and presentation intent, while the framework shares the machinery for rendering, synchronization, storage, and agent access.

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Dimension	Component Frameworks	Low-Code / Visual Builders	Strvct
Primary presentation artifact	Components and application composition	Visual layouts and bindings	Annotated nodes and presentation policies
Reuse mechanism	Shared components, libraries, schema-driven tools	Platform widgets, templates, connectors	Shared views interpreting node and slot metadata
Presentation decisions	Usually expressed in components	Usually expressed in visual configuration	Expressed in model metadata, node methods, and presentation nodes
Maintenance trade-off	App authors maintain composition and integration	App authors maintain layouts and platform configuration	Shared mechanics reduce repeated view code; metadata and domain complexity still grow
Agent integration	Application- or library-defined APIs and tools	Platform- or application-defined connectors	Schema-and-patch operations derived from exposed metadata
Persistence	Chosen and integrated by the application	Platform-dependent	Framework object pools and cloud synchronization

Table 3. Architectural defaults of UI development paradigms.

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